

No.	Solution
7(a)	charge distribution A very small number of alpha particles directed towards the gold foil is deflected by very large angles from their original paths, indicating that the positive charge in the atom is not uniformly distributed, but concentrated within a small volume of space in the atom. mass distribution A very small fraction of the alpha particles are deflected by 180°, indicating that all the positive charges and mass of the gold atom are concentrated in a very small volume within the atom which has sufficient mass to cause the alpha particles to recoil.
7(b)	Loss in KE of alpha particle = Gain in EPE of alpha particle $\frac{1}{2}mv^2 = \frac{Q_{\text{Au}}Q_{\text{He}}}{4\pi\epsilon_0 r}$ $5.59 \times 10^6 \times 1.60 \times 10^{-19} = \frac{(79)(1.60 \times 10^{-19})(2)(1.60 \times 10^{-19})}{4\pi \times 8.85 \times 10^{-12} r}$ $r = 4.066 \times 10^{-14}$ $\approx 4.07 \times 10^{-14} \text{ m}$
7(c)	Nuclear reaction ${}_{80}^{222}\text{Rn} \rightarrow {}_{78}^{218}\text{Po} + {}_2^4\text{He} + \underset{\text{energy}}{\gamma}$ Energy released = Binding energy of products – Binding energy of reactants $6.62 = \text{BE}_{\text{Po}} \times 218 + (7.08 \times 4) - (7.69 \times 222)$ $\text{BE}_{\text{Po}} = 7.73 \text{ MeV}$
8(a)	The X-ray tube is evacuated so that the electrons can move from the filament to